BST Strong-Sector Package — Introduction (v0.3, forces-first, boundaries-clear). Built to spec (Casey, K1716). FORCES (lead): geometry forces the color NUMBER N_c=3, the real-type SO(3), and baryon number; the asymptotic-freedom SIGN β_0>0 lands blind from the a_2 heat-kernel coefficient ($\mathbb{R}^4$-valid, local); m_p/m_e = $6\pi^5$ = 1836.12 (0.002%). BOUNDARIES (page one and close): the compact-boundary '6/a^2' is a Kaluza–Klein gap, NOT the Clay mass gap (opposite scaling); the SU(3) gauge DYNAMICS is imported, not derived (#108); the Koide relation is a labeled patchwork, not a single-mechanism derivation; n_C=5 is a measured boundary (an input), not forced. ORGANIZING PRINCIPLE (K1716): local heat-kernel coefficients are geometry-given ($\mathbb{R}^4$-valid); non-local / compactness-dependent quantities are physics-hosted — the geometry-vs-physics seam as a checkable test. Emerging frontier (being computed, NOT yet claimed): the a-theorem central charges a, c as local

coefficients from the same validated bracket. v0.3 draft; ships on Casey GO + Keeper-PASS + Cal-vet + both voices.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace, Elie, Cal, Keeper

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The Strong Sector from One Geometry

What D_IV^5 forces, what it hosts, and the seam between them

Two-paper package — Paper A (the strong sector: forced color number, blind one-loop running, a compactification gap that is not the Clay gap) · Paper B (D_IV^5 substrate uniqueness).

The forces (what geometry decides)

From one bounded symmetric domain — D_IV^5 = $SO_0(5,2)/[SO(5)\times SO(2)]$, forced (not chosen) by criteria-innocent uniqueness — the strong sector's *identity* falls out with no adjustable dimensionless numbers:

- The color number is forced. N_c = 3, the real-type SO(3), and baryon number are outputs of the substrate (Paper B: the scalar-Wallach equality N_c = rank$^2-1$ = 3, via the proved T1829) — not inputs.
- Asymptotic freedom lands blind. The one-loop β-function *sign*, $\beta_0 > 0$, comes out of the a_2 heat-kernel coefficient of the fielded gauge operator with nothing imported — the paramagnetic (spin) contribution beating the diamagnetic (orbital). This is $\mathbb{R}^4$-valid (a short-distance, *local* result), and it is the genuine dynamical credential of the package.
- The proton weighs what it does. m_p/m_e = $6\pi^5$ = 1836.12 against the measured 1836.15 — 0.002% (T187), from rep-theory (N_c!·π^{n_C}), no fit.

These are not coincidences dressed as derivations. They are *local, geometry-given* facts, and the section below explains — with a checkable test — why we trust them and where we draw the line.

The organizing principle: the geometry–physics seam (K1716)

BST forces the container; it hosts the dynamics. The seam between them is not a matter of taste — it is a checkable test:

Local heat-kernel (Seeley–DeWitt) coefficients are geometry-given and $\mathbb{R}^4$-valid. Non-local, or compactness-dependent, quantities are physics the container hosts — not derived by the geometry.

- Local ⇒ $\mathbb{R}^4$-valid (a force): the a_2 coefficient (asymptotic-freedom sign) is a local curvature invariant; it survives to flat space. *Trusted.*
- Compactness-dependent ⇒ hosted (a boundary): a spectral gap of a free operator on the compact boundary scales as $1/a^2$ and vanishes as the space decompactifies ($a \to \infty$). That is Kaluza–Klein kinematics, not an interacting mass gap. *Labeled, not claimed.*

This one test both *earns* the forces above and *draws* the boundaries below. It is why the package is honest by construction rather than by apology.

The boundaries (stated here, on page one, and again at the close)

Four things geometry does not give, said plainly so no reader and no referee has to find them:

1. The compactification gap is not the Clay mass gap. The boundary eigenvalue 2(d_S−1) = 6 (dimension-rooted: d_S = n_C−1 = 4; U(1)/SU(3)/SU(5) all give 6 — the group sets degeneracy, not the eigenvalue) is a free-Hodge / Kaluza–Klein gap, $6/a^2 \to 0$ as $a \to \infty$. The Clay Yang–Mills gap is interacting and fixed in that limit. Opposite scaling is a contradiction, not an unbuilt bridge — the Clay mass gap is left open (K940). We never say "we derived the Yang–Mills mass gap."
2. The SU(3) gauge *dynamics* is imported. Geometry forces the color *number* and its SO(3)/baryon structure; the promotion to a gauged non-abelian Yang–Mills theory (the gauge principle / inner fluctuation) is physics the container hosts (#108, K1677). We never say "the gauge group is derived."
3. The Koide relation is a labeled patchwork. The charged-lepton Koide phase is fitted-region structure assembled from parts, not a single forced mechanism; we present it as a patchwork and mark it so — no clean-derivation claim.
4. n_C = 5 is a measured boundary (an input). The complex dimension 5 is fixed by four independent routes as the *measured* boundary of the construction (K1697), not forced from nothing. It is the one place we take an input beyond the ruler, and we state it as an input, in the GR/QM manner.

The bright-high-schooler version

The strong nuclear force usually gets three facts *fed in*: why it has three "colors," why it gets weaker at very short distances (asymptotic freedom — a Nobel discovery), and how heavy the proton is. This package gets all three *out* of one shape, D_IV^5 — the color number 3, the sign of the short-distance weakening, and the proton-to-electron mass ratio to four decimals.

The honest part, said first: the same shape has a "smallest vibration," like a finite drum's lowest note, and it comes out a clean 6. But that note is just the drum being *finite* — make the drum infinite and the note drops to zero — whereas the real Yang–Mills mass gap *stays*. They behave oppositely, so ours is not that famous gap, and we don't pretend it is. The rule that tells us which of our results to trust is simple: anything that survives making the shape infinitely large is real physics; anything that needs the shape to be finite is just the shape talking. Asymptotic freedom survives. The gap does not. We keep the first and label the second.

An emerging frontier (noted, not claimed)

The same validated a_2 bracket whose curvature-squared half gave asymptotic freedom also carries the trace anomaly — the central charges a and c, and the a-theorem (the monotonicity of a under renormalization flow). Because these are local coefficients, the seam test says they are $\mathbb{R}^4$-valid. As a check, a free conformal scalar gives a : c = 1 : 3, the standard value. Computing BST's a and c blind from the bracket is the strong sector's honest next frontier — a *credential* (BST computes standard-QFT central charges as local coefficients), with the monotonicity/a-theorem line held separate until earned. This is in progress and is not part of the package's claims.

Reading order

Read Paper B first (why D_IV^5 and N_c = 3 are forced), then Paper A (the running, the KK gap, and the Clay gap left open). Both restate the seam test; Paper A's Section 0 puts the KK-vs-Clay scaling up front.

Close — the boundaries, restated

The package's claim is exactly this: *forced geometry gives the color number (3, SO(3), baryon#), the sign of the running coupling, and the proton mass ratio — all local, all $\mathbb{R}^4$-valid — while the Yang–Mills gauge dynamics is imported, the compactification gap is not the Clay mass gap, the Koide relation is a patchwork, and n_C = 5 is a measured input.* One geometry, one ruler, zero chosen dimensionless numbers among the forces — and every boundary named twice, here and on Paper A's first page. Never "we solved a Millennium Problem"; never "the gauge group is derived"; never a mass-gap claim.

Package intro v0.3 (forces-first, boundaries-clear; supersedes v0.2). Built to Casey's spec (K1716). Paper A = v1.1 RE-SCOPED (K1714/K1715). Paper B = v0.6. Organizing principle: the local($\mathbb{R}^4$-valid)/non-local(hosted) seam. a-theorem = emerging frontier, not claimed. Ships on Casey GO + Keeper-PASS + Cal-vet + both voices. Nothing pushed; CP existence-only.

— Lyra, Wednesday 2026-08-19 (date-verified).